

Ghazal Lahooti

Urban Designer & Researcher

Nature-based solutions & climate adaptation · Data-driven & computational design · Evidence for early design decisions

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Portfolio · *Click or Scan*

Profile

Urban designer and researcher with a background in architecture, working on urban heat and nature-based solutions (NBS). My research asks *where* cooling should be prioritised, based on where heat meets people's everyday routes, and *which* NBS each street can accommodate. I combine remote sensing, GIS, sun and wind simulation and parametric design, and I am seeking a PhD or research assistant role in nature-based climate adaptation, particularly in data-driven and AI-assisted methods that turn evidence on NBS into early design decisions.

Research interests: nature-based solutions & climate adaptation · urban heat & outdoor comfort · data-driven & AI-assisted design · healthy ageing in public space
Core tools: Rhino & Grasshopper · Ladybug · Eddy3D & SimScale · Galapagos · QGIS · Google Earth Engine · Python

Education

- **M.Sc. in Sustainable Urban Design**, Lund University (LTH), Sweden Sep. 2024 – Oct. 2026
  Thesis: *A Context-Specific Nature-Based Framework for Heat Mitigation: Supporting the Everyday Mobility of Older Adults in Kadıköy, Istanbul*
  Supervisor: Teresa Arana Aristi · Examiner: Andrew Karvonen
- **B.Sc. in Architectural Engineering**, Iran University of Science and Technology, Iran Sep. 2018 – Jul. 2023
  GPA: 16.95/20 · Thesis: *Sustainable Mountain Refuge at the Third Campsite of Mount Damavand*

Research Experience

- **Research Intern**, Institute for Design Strategies, TH OWL, Detmold, Germany Jun. – Aug. 2025 & Oct. – Dec. 2026
  AIM-NBS: Advancing Impact Measurement of Nature-Based Solutions
  - Phase 1: covered 6 of 12 societal challenges in a PRISMA umbrella review of NBS impacts (410 papers); synthesised impact indicators and visualised results in Python.
  - Phase 2: recalibrating the review findings for the final synthesis and co-writing the project report (credited).
- **Master's Thesis Research**, Lund University, fieldwork in Istanbul Jan. – Oct. 2026
  How can NBS cool older adults' everyday walks in Kadıköy, Istanbul's district with the most residents aged 65+?
  - Mapped ten summers of land surface temperature (Landsat, Earth Engine) against residents aged 65+.
  - Linked 124 everyday destinations into the *Green Loop*, a cooling network within 100 m of two-thirds of them.
  - Grouped 10 streets into 4 typologies with Ladybug sun simulation; designed with a 7-tool NBS toolbox.
  - **Outcome:** a transferable framework: social context decides *where* to cool, physical context *how*.
- **Project Assistant**, Lund University & City of Malmö Oct. – Nov. 2025
  Microclimate analysis of Hyllie, Malmö, Sweden
  - Rebuilt the 3D model for a CFD wind study and mapped direct sun hours per grid cell.

Tool Development

- **UrbanSim: heat-screening concept prototype** (Founder & sole developer) Summer 2026 – present
  Sustainability Leapfrogs, LU Innovation, Lund University
  - Browser-based concept (JavaScript, OpenStreetMap, Landsat) for ranking streets by sun exposure, destinations and surface heat, and matching them to NBS via source-cited rules.
  - **Status:** conceptual MVP, not yet validated with real data. Next: an evidence-graded NBS library.

Research Funding & Awards

- **Sustainability Leapfrogs Scholarship**, LU Innovation (SEK 40,000), 2026: development of UrbanSim
- **Crafoord Foundation Travel Grant** (SEK 30,000), 2026: thesis fieldwork in Istanbul
- **Lund University Global Scholarship** (75% tuition waiver), 2024–2026: awarded for academic merit

Research Outputs

- Lahooti, G. (2026). *A Context-Specific Nature-Based Framework for Heat Mitigation: Supporting the Everyday Mobility of Older Adults in Kadıköy, Istanbul.* M.Sc. thesis, Lund University.
- AIM-NBS project report, Institute for Design Strategies, TH OWL (in preparation; credited contributor).

Professional Experience

- **Junior Architect**, Cedrus Architecture Studio, Tehran Jan. 2023 – Aug. 2024
  - Site and environmental analyses and precedent research for early-stage housing and building design.
  - Prepared presentation documents, visualisations, animations and physical models for design proposals.

Skills & Methods

- **Research methods:** PRISMA systematic & umbrella reviews, interviews, street surveys, demographic analysis
- **Spatial data:** QGIS, Google Earth Engine, Python (GeoPandas, Pandas, NumPy, Matplotlib), OpenStreetMap
- **Simulation & computational design:** Rhino, Grasshopper, Ladybug, Eddy3D, SimScale, Galapagos
- **Programming:** Python, JavaScript
- **Design & visualisation:** Revit, AutoCAD, SketchUp, Blender, Enscape, D5, Adobe Creative Suite

Languages

Persian: Native · **English:** C1 (IELTS Academic 7.5) · **German:** B1 (Goethe-Zertifikat) · **Swedish:** A1, learning

References

- **Available Upon Request**